

FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY

DEPARTMENT OF ELECTRICAL, ELECTRONIC AND COMPUTER ENGINEERING

EBN LABORATORY SAFETY APPENDIX

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I. LABORATORY SAFETY

The Occupational Health and Safety Act (Act 85 of 1993, as amended by Occupational Health and Safety Amendment Act, No. 181 of 1993) stipulates:

"To provide for the health and safety of persons at work and for the health and safety of persons in connection with the use of plant and machinery; the protection of persons other than persons at work against hazards to health and safety arising out of or in connection with the activities of persons at work".

By law, laboratory facilities should provide a safe environment for students to work in, but at the same time, students working on projects should not create a hazardous work environment for other students.

Any student using any of the undergraduate labs of the Department of Electrical, Electronic and Computer engineering, should please take note of the following usage rules. These are strictly enforced.

A. Lab Rules

Record is kept of offenders.

First offence: access to all departmental laboratories will be cancelled for **one week** if any of these rules are not adhered to.

Second offence: The case will be referred to UP Legal Services for disciplinary action, and the student will be expelled from all departmental laboratories at least **until the outcome of a disciplinary hearing is known**.

All these rules apply 24/7.

- **Food in the labs**: No food or liquids are allowed in the laboratories at any time.
- **Music and video in the labs**: No radios or other music reproducing devices are allowed in any Departmental laboratories at any time, whether with or without earphones, the viewing of any video content is prohibited, unless part of a formal class.
- **Loud talking or other noise in the labs**: The laboratory is a learning environment and students will not be allowed to disturb others. The laboratories are not social venues and may not be used as such. No loud talking or any other noise or disturbance to fellow students is allowed.
- **Behavior in the labs**: Students are expected to act professionally. No unruly behavior will be tolerated in the laboratories.
- **Clothing in the labs**: For security reasons, no caps or hats (including Beanies) may be worn in the laboratories.
- **Student cards**: For security reasons, student cards must be worn in the labs always (attached to clothing above the waist) and must clearly visible. Students are obliged to produce their student card on the request of any member of personnel.
- **Shoes**: Closed shoes must be worn in ALL laboratories always.
- **Instructions by a lab instructor**: students are obliged to carry out any instructions from a lab instructor (unless these violate any of the other rules or could potentially create an unsafe situation). If you are instructed to leave the lab, do so immediately.
- **Signs**: Take note of the "prohibited" signs in the laboratory.
- **Lab usage**:

- After use, resistors and capacitors must be returned to their correct designated trays.
- Components that had been distributed, must be handed in at the end of the practical.
- Connecting wire must also be returned.
- Soldering irons without proper stands are prohibited.
- Benches must be switched off at the end of the practical.
- Chairs should be pushed in under the bench after use.
- Tidy up your bench after use (remove all papers, cut-off wires etc.).
- Defective equipment must be reported to the person in charge of the specific laboratory, with a description of the problem including your contact details should further information be needed. Remember to include the lab and bench number!
- **Note: Laboratories are intended for module-related experimental work where lab instruments are needed, and not for ANY other use. No labs may be used as a study center.**

Complaints procedure. Students that wish to report incidents or non-compliance with these rules, should send an email or report incidents personally to Mr Helmut Schütte (Lab A, Lab B, Lab C and Project Lab 1) or Mr Willem van Jaarsveld (NetLab A, NetLab B, CAEC and Project Lab 2) or Mr Mickey Marimuthu (Lab M) as soon as possible, but not more than 24 hours after the incident has occurred. Provide detail such as incident, time, place, and name of students involved where available.

B. Electrical Safety

1. Know the location of the following safety items within the lab:

- a. Fire extinguisher
- b. First aid kit
- c. Telephone and emergency numbers: Make sure emergency phone numbers are close at hand to call for assistance if necessary. The extension for emergencies is 2310 or (012) 420 2310. If any safety questions arise, consult the lab instructor or staff for guidance and instructions. The 24-hour crisis service line is 0800 00 64 28 (Toll free).

Observing proper safety precautions is important when working in the laboratory to prevent injuries to one self or others. Familiarise yourself with the operation of the fire extinguishers in the lab.

Fire extinguisher ratings are as follow:

- A: Burning paper, wood, rubbish or textiles.
- B: Flammable liquid.
- C: **Electrical Fires.**

2. Electric shock

Electric shock is caused when an electric current passed through the human body. The severity depends mainly on the amount of current and is less a function of the applied voltage. The threshold of electric shock is about 1 mA which usually gives an unpleasant

tingling. For currents above 10 mA, severe muscle pain occurs, and the victim can't let go of the conductor due to muscle spasm. Current between 100 mA and 200 mA (50 Hz AC) causes ventricular fibrillation of the heart and is most likely to be lethal unless medical help is available within minutes. Best is to phone an Ambulance.

What is the voltage required for a fatal current to flow? This depends on the skin resistance. Current is dependent on the applied Voltage and the resistance of the circuit. Wet skin can have a resistance as low as 150 Ω and dry skin may have a resistance of 15 k Ω . Arms and legs have a resistance of about 100 Ω , and the torso, 200 Ω .

This implies that 220 V (the standard voltage of South African mains electricity) can easily cause more than 200 mA to flow in the body if the skin is wet and thus be fatal. In addition, skin resistance falls quickly at the point of contact the longer it remains in contact with the electric circuit, so it is important to break the contact as quickly as possible to prevent the current from rising to lethal levels.

The following sections describe several safety precautions, which one should always follow.

3. General Safety Precautions

The following points describe general safety precautions within the laboratory and should be followed always:

- a. Do not work alone while working with high voltages or if using electrically operated machinery, like a drill.
- b. Never leave high-voltage appliances on when not present.
- c. Make sure all high voltage connections are adequately taped or otherwise insulated to prevent any accidental contact.
- d. After switching the power off, discharge any capacitors that were in the circuit. Do not trust supposedly discharged capacitors. Certain types of capacitors can build up a residual charge after being discharged. Use a shorting bar across the capacitor, and keep it connected until ready for use.
- e. While using electrolytic capacitors, do not put excessive voltage across them or connect them in reverse polarity or connect AC across them (this could easily break the component).
- f. Take extreme care using tools (such as side-cutters) that can cause short circuits if accidental contact is made to other circuit elements. Only tools with insulated handles should be used.
- g. If a person comes in contact with a high voltage, immediately shut off the power. Do not attempt to remove a person in contact with a high voltage unless you are insulated.
- h. In the event of an electrical fire, do not use water to extinguish it. The lab fire extinguishers are specifically designed for electrical fires. Vacate the lab and close the door. Avoid breathing in toxic smoke or fumes. Activate the fire alarm if possible.
- i. Check wire current carrying capacity if high currents will be used. The lab power wiring can only conduct 15 Amperes continuously.
- j. Make sure all leads are rated to withstand the voltages be used. This includes instrument leads. Common wire insulation is rated for 600 Volts.

- k. Avoid simultaneous touching of any metal chassis used as an enclosure for circuits and any pipes in the laboratory that may make contact with the earth, such as a water pipe. Use a floating voltmeter to measure the voltage from ground to the chassis to see if a hazardous potential difference exists.
- l. Make sure that the lab instruments are at ground potential by using the ground terminal supplied on the instrument.

4. Equipment grounding

Grounding is very important. Improper grounding can be the source of errors, noise and safety hazards.

Electric instruments and appliances have equipment cases that are electrically isolated from the wires that carry the power. The isolation is provided by the insulation of the wires. However, if the wire insulation is damaged and makes contact with the case, the case will be at the high voltage supplied by the wires. If the user touches the instrument he or she will feel the high voltage. If, while standing on a wet floor, a user simultaneously comes in contact with the instrument case and a pipe or faucet connected to ground, a sizable current can flow through him or her. However, if the case is connected to the ground by use of a third (ground) wire; the current will flow from the hot wire directly to the ground and bypass the user.

Equipment with a three-wire cord is thus much safer to use. The ground wire (3rd wire) which is connected to metal case is also connected to the earth ground (usually a pipe or bar in the ground) through the wall plug outlet.

Grounding may also prevent static electrical discharge, which could ignite flammable materials or damage components and equipment.

C. Emergency Procedures

The following procedures should be taken note of in the case of an emergency. Staff and students are again reminded of the general UP landline and cell phone emergency numbers, namely (012) 4202310 or 0836540476. Staff and students are encouraged to save these numbers in their contacts lists for reporting any disruption or emergency.

1. Evacuation procedure:
 - a. An alarm is activated. Alert other individuals as well.
 - b. Follow the instructions of the occupational health and safety representative.
 - c. Follow the direction pointers to the nearest emergency exit.
 - d. Walk briskly but do not run.
 - e. Help the handicapped and people who are struggling.
 - f. Once outside, proceed to the nearest assembly point.
 - g. Report to the occupational health and safety representative of your area for the roll-call.
 - h. Wait at the assembly point for further instructions.
2. Fire incident:
 - a. Sound the alarm immediately and notify the Security Services at 012 420 2310.
 - b. Attempt to extinguish the fire with available fire-fighting equipment.

- c. Evacuate if the fire cannot be extinguished according to procedure 1.
 - d. Close all doors and windows.
 - e. Lock away or take with you all classified documents or personal belongings.
 - f. Avoid smoke-filled areas and stay close to the floor where the air will be less toxic.
 - g. Shout at regular intervals to alert emergency crews of your location.
3. General hints:
- a. Always stay calm.
 - b. Make sure you know where the location of the emergency equipment is and how to use it.
 - c. Familiarise yourself with the escape route.
 - d. Assist visitors who are not conversant with the emergency procedures.
 - e. Be prepared to identify and report foreign objects and suspicious persons.
 - f. No information may be given to the news. Media or any other organisation not involved.
4. OHS contacts
- a. Ludolf van der Merwe: Deputy Director Maintenance and Operations:
 - (012) - 4202079 ludolf.vandermerwe@up.ac.za
 - b. Yolanda Bekeur: OHS Manager:
 - (012)-4203308 yolanda.bekeur@up.ac.za
 - c. Sonja Sauer: Occupational Health Nursing Practitioner (OHNP):
 - (012) - 4204409 sonia.sauer@up.ac.za
 - d. Bobby Hand: Senior Fire and OHS Coordinator:
 - (012) - 4203305 bobby.hand@up.ac.za
 - e. Cynthia Mamatela: Radiation Control Officer:
 - (012) - 4206322 cynthia.mamatela@up.ac.za
 - f. Carol Mbele: OHS Coordinator:
 - (012) - 4203332 carol.mbele@up.ac.za
 - g. Gift Nkosi: Senior Fire Equipment Inspector:
 - (012) - 4204088 gift.nkosi@up.ac.za
 - h. Hennie Venter: Inspector-lecture halls:
 - (012) - 4206968 Hendrik.venter@up.ac.za

II. FIRST AID

In all cases of laboratory emergencies, the lab instructor should be called as soon as the victim(s) are attended to. The instructors for the labs are:

- Mr Helmut Schütte (Lab A, Lab B, Lab C and Project Lab 1).
- Mr Willem van Jaarsveld (NetLab A, NetLab B, CAEC and Project Lab 2).
- Mr Mickey Marimuthu (Lab M).

A. Electrical Burns

1. Electrical burns should be handled differently regarding all other burns.
2. With electrical burns, extreme caution must be taken regarding your own safety.

3. Power must first be switched off at the main switch including power at the 'plug'. The contact wire must not be removed with a non-conducting source before the power has been isolated.
4. The current that goes through the body can cause irregular function of the electrical conduction system of the heart or can stop the victim's heart (CPR must be started with high success rate).
5. With electrical burns, it is important to keep in mind the possibilities of fractures, or spinal injuries, the patient must be treated medically, and the burns should be dressed properly for journeys to the hospital.
6. Electrical burns cause devastating internal injuries with little external evidence.
7. Burn wounds can be covered with a sterile or clean dressing.

B. Electrical Shock

If electric shock occurs, the victim may not be able to break contact with the circuit. Contact between the victim and the circuit must be broken as quickly as possible.

1. Do not touch the victim with any part of your body or with any material that can conduct electricity.
2. Turn off the power to the circuit from the main switch, not just at the plug or source of contact.
3. If power cannot be turned off, contact between the victim and the circuit must be broken with a non-conducting material, such as a broom.
4. Find help immediately and contact one of the lab instructors on duty.